

Assignment 1

Goal

The goal of this assignment is to give you hands-on experience with evaluating, analyzing, and improving large language models (LLMs) on a task: ***translating natural-language questions into SQL (Text-to-SQL)***. You will (1) benchmark LLMs of different sizes and types, (2) analyze the reasoning traces produced by reasoning-capable LLM(s) and relate them to task performance, and (3) improve one of your implemented solutions either by improving its reasoning or by adapting the model's parameters.

Warning: Start early. Running LLMs on hundreds of examples, especially reasoning models, can take hours, and free GPU sessions (e.g., Google Colab) have time limits. Improving the model would also need time and multiple trials.

Dataset

You will use the **Spider 1.0** dataset (<https://github.com/taoyds/spider>), a large-scale, cross-domain Text-to-SQL benchmark. Each example consists of a natural-language question, the schema of the target database (SQLite), and a gold SQL query.

- Use the training split only for few-shot example selection (Task 1/3) or for model improvement (e.g., fine-tuning, Task 3).
- Use the development split for all evaluation.
- Evaluation (sub)set: Due to computing resource limits, you may choose to evaluate on a subset of the development split. If you only use a subset, ensure your sampling strategy would give you a meaningful sample set and record the list of example IDs you used.
- Metrics: You need to decide which metric(s) to use, by default, follow the paper of this dataset.
- For difficulty of tasks, you could use the official evaluation script (<https://github.com/taoyds/test-suite-sql-eval/blob/master/evaluation.py>).

Submission Package

You are required to submit two items:

1. **Final Code:** Your Jupyter notebook(s) or scripts for replicating your experiments (Task 1-3). You should ensure the code is executable (assuming the TA could download/use the same LLM) and include a `README` with setup instructions. The TA may run your code on the evaluation subset. Do not submit model, do not reveal your API token.

2. **Written Report (PDF):** In the report, write required answer to four tasks following the below report requirements.

Report Requirements

The report is the main basis for grading. All results, settings, and required specifications must appear in the report itself. **TAs will not search your notebooks or code to find results or missing details, and anything absent from the report will be treated as not done.** The code is for replicability check (not passing this check would lead to huge deduction of scores).

Organize your report using exactly the following headings:

1. Task 1 – RQ1: Methodology; Results
2. Task 2 – RQ2: Methodology; Results
3. Task 3 – RQ3: Methodology; Results
4. Task 4 – AI Usage Declaration

Under each Methodology heading, report every item listed for that task, in the same order. Every table and figure must have a caption and be referenced in the text.

Task 1: Benchmark LLMs of Different Sizes and Architectures

Implement a Text-to-SQL pipeline and evaluate **at least three models** that together cover the following two categories:

Category	Requirement	Example candidates (not mandatory)
A. Decoder-only LLMs (no explicit reasoning)	At least 2 models of different sizes from the same family	Qwen 2.5 or Qwen 2.5-Coder
B. Reasoning LLMs	At least 1 model that outputs an explicit reasoning trace	DeepSeek-R1-Distill-Qwen, Qwen3+ with thinking mode enabled

You may use open-weight models (e.g., via Hugging Face on Google Colab or your own other compute resources) or API-based models.

Report (Task 1): Write up Task 1 as the Methodology and Results sections that answer the following research question:

RQ1: How do (selected) LLMs perform on cross-domain Text-to-SQL?

Methodology. Describe your study in enough detail that another people could reproduce it. At a minimum, report:

- Models: name, version or checkpoint, parameter count (for mixture-of-experts models, give both total and active parameters), quantization, and how you accessed the model (local inference or API).
- Evaluation data: the Spider split you used, how you sampled the subset, and the random seed.
- Prompting and input: the prompt template, how you represented the database schema, any few-shot examples, and how you extracted the final SQL from each model's output.
- Inference settings: decoding strategy, temperature, top-p, maximum output length, whether reasoning was enabled, and the hardware or runtime used.
- Evaluation: your metrics and how you counted invalid or unparseable outputs.

Results. Report performances of all considered models, overall and by difficulty level, in a summary table. Also report efficiency measures (e.g., inference time and output tokens per example). Present results first, then interpret them: answer RQ1 directly and support each claim with specific numbers from your table rather than general impressions.

Task 2: Analyze Reasoning Traces

Report (Task 2): Write up Task 2 as the Methodology and Results sections that answer the following research question:

RQ2: How do characteristics of reasoning traces (of reasoning LLM) relate to Text-to-SQL performance?

This RQ is based on the reasoning traces collected from the reasoning LLM used in Task 1.

Methodology. Define at least two measurable features and extract them from every trace. For each feature, give its definition, why it is relevant to Text-to-SQL, and how you computed it.

Name the statistical tests or models you used to relate each feature to correctness and difficulty. Explain why each fits your data.

Results. Report descriptive statistics for each feature, broken down by correctness and difficulty. Add the statistical results and at least one visualization, for example box plots of each feature by correctness and difficulty.

Task 3: Improve One Solution

Report (Task 3): Write up Task 3 as the Methodology and Results sections that answer the following research question:

RQ3: To what extent does [strategy] improve [model]?

Select one model/solution from Task 1 and improve its performance. Choose **one** of the following options:

- **Option A** - Improving Reasoning. Implement a strategy that may improve how the model reasons about the question and schema.
- **Option B** - Adaptation / Learning to Adapt (model parameters ARE changed). Implement a strategy that may improve model's capabilities on this specific task/domain. Adapt the model's parameters using the Spider training split.

Methodology. Describe which model/solution you consider for task 3 and justify why it is interesting to improve this approach.

Describe your improved solution, motivate it, name it, and replace [strategy] with this name in RQ3.

Evaluate your improved solution it on the same evaluation subset and metrics as Task 1, overall and by difficulty level, and compare against the original solution.

A negative result is acceptable if it is well motivated, correctly evaluated, and properly explained.

Results. Report performances for the original solution, the improved solution in a single table, overall and by difficulty level. State whether your expectation from the motivation held, supported by specific numbers. Illustrate with one or two examples that the improvement fixed or broke.

Task 4: AI Usage Declaration

At the end of your report, write one paragraph declaring how you used generative AI tools (e.g., ChatGPT, Claude, GitHub Copilot) in this assignment. State which tools you used, which parts of the work they supported (e.g., writing or debugging code, designing prompts, choosing statistical tests, or editing the report), and how you checked the output they produced. If you did not use any generative AI tools, state that explicitly. The LLMs you evaluated as part of the tasks do not need to be declared here. Your declaration does not affect your grade unless it is inconsistent with the course's academic integrity policy. You remain fully responsible for the correctness of your code, results, and claims.